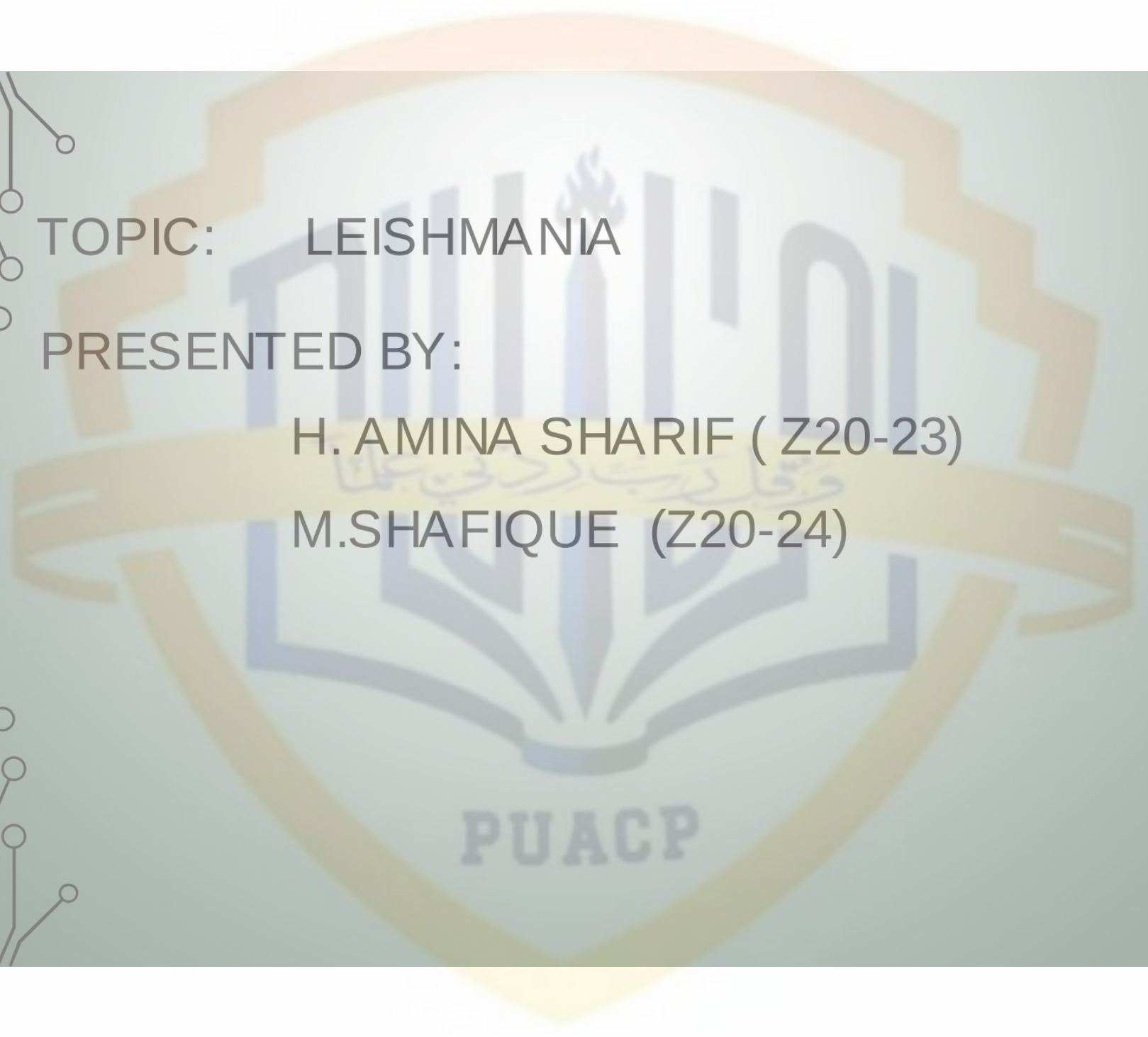




TOPIC: LEISHMANIA

PRESENTED BY:

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INTRODUCTION

- Leishmania is a flagellate parasite of man and other mammals, especially dogs and rodents.
- Leishmania causes different kinds of infections collectively referred to as leishmaniasis.
- It is spread by the bite of sandflies of genus phlebotomus in the new world and of genus lutzomyia in the new world.
- It caused the disease leishmaniasis which is 2nd largest parasitic killer in the world.
- It is world wide in distribution.

CLASSIFICATION

- Kingdom Protista
- Phylum protozoa
- Class zoomastigophora
- Order kinetoplastida
- Genus Leishmania

MORPHOLOGY

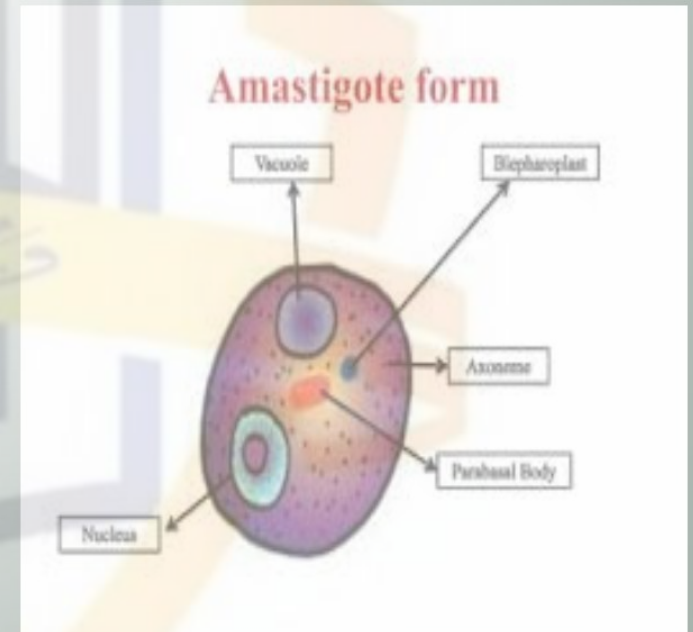
➤ The parasite exists in two forms

I. Amastigotes

➤ It is an a-flagellar stage that occurs in vertebrate host.

➤ The shape is round or oval with 2-4 micrometers along longitudinal axis

➤ Divides by binary fission and has a larger nucleus which is situated centrally



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Promastigotes

- It is a flagellar stage occur in the sand fly.
- It has spindle shape with 15-20 micrometer in length and 1-2 micrometer in width.
- Nucleus is smaller in size and situated in the middle of the cell or along the side of the cell wall.



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HABITAT

- These are essentially the parasites of visceral organs.
- Promastigote forms found in sand fly and in culture.
- Amastigote forms found in man in reticuloendothelial cells of spleen, bone marrow, liver, intestinal mucosa, mesenteric lymph node.

VECTOR S



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TYPES

➤ Leishmaniasis is divided into clinical syndromes according to what part of the body is affected most.

1. Visceral leishmaniasis
2. Cutaneous leishmaniasis
3. Mucocutaneous leishmaniasis

VISCERAL LEISHMANIASIS

- It is also known as kala-azar.
- It is caused by leishmania L.donovani.
- Part of the body affected most is internal organ.

SYMPTOMS

- Malaise
- Skin darkening
- Enlargement of spleen
- anorexia



CUTANEOUS LEISHMANIASIS

- It is the most common type of leishmaniasis.
- It is caused *L. mexicana* and *L. braziliensis*.
- This is the most severe and chronic type.
- This infection produces a skin ulcer which leaves an unsightly scar on healing.



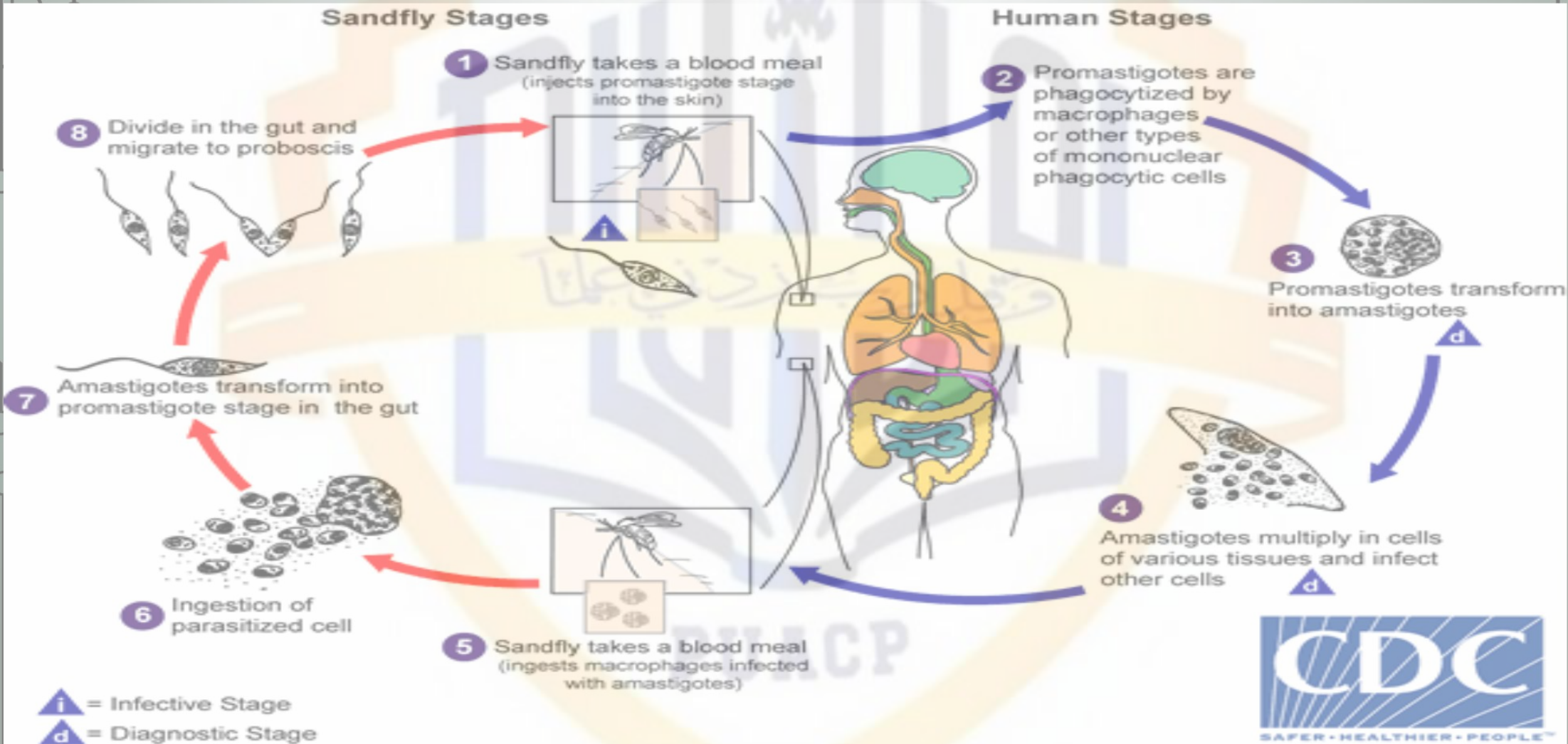
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MUCOCUTANEOUS LEISHMANIASIS

- This is also known as espundia.
- This is caused by *L.braziliensis* and occasionally by *L. panamensis*.
- This initially produces lesions like those of cutaneous but later often after 30 years metastatic spread may occur into oronasal and pharyngeal mucosa, resulting in highly disfiguring tissue damage.
- Part of the body affected most is skin and mucous membrane of nose and pharynx.



LIFE CYCLE



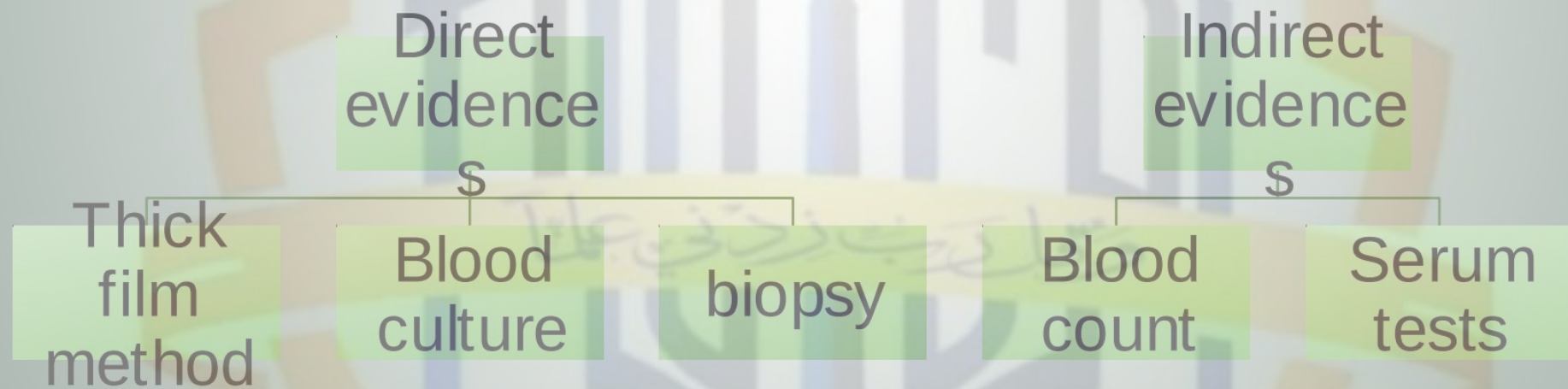
MODE OF TRANSMISSION

- Mainly by bite of sand fly (vector).
- Less frequently by
 - I. Blood transfusion
 - II. Congenital infection
 - III. Accidental inoculation of cultured promastigotes in lab.
 - IV. Sexual intercourse

CLINICAL MANIFESTATIONS

- Fever
- Spleen enlargement
- Lymphadenopathy
- Darkening of skin
- Complications like pneumonia, TB, dysentery, uncontrolled haemorrhage.

LABORATORY DIAGNOSIS



PREVENTION AND CONTROL

Reduction of sand fly population

- By insecticides mainly DDT, dieldrin, malathion.

Reduction of reservoir

- By killing infected dogs in the cases of zoonotic kala-azar.

Education in the community

- About the causes and modes of transmission of leishmaniasis.

Prevention of exposure to sand fly

- Using insect repellent, bed nets and window mesh as needed.

TREATMENT

DRUGS

Sodium
stibogluconat
e solution

Amphotericin
B

Pentamidine

Mitlefosine

interferon

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*Thank
You*

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Sege and Forest Green Centre Cam by ananabkym

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